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Comparative Analysis of Multi-Agent AI Systems in Healthcare

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Abstract

Multi-Agent System (MAS) has emerged as an important framework for artificial intelligence in healthcare applications. It enables independent agents to work together in complex healthcare workflows. Unlike traditional AI models, MAS models break down clinical problem solving into independent agents that communicate more effectively while making decisions and addressing issues. Each agent is optimized for a specific task, including data collection, diagnosis, and ethical oversight. As the global healthcare system faces more long-term diseases and challenges in critical care, a smart multi-agent architecture offers a promising way to support real-time, patient-focused healthcare.

Despite significant advances in MAS research, the framework has many limitations, such as the lack of universally accepted standards for comparing MAS frameworks. This absence of standards makes it difficult to compare or interpret different implementations. The lack of consistency has led to varying approaches to key aspects of the systems. Important factors like privacy preservation, ethical governance, transparency, and reliability are often addressed separately across research. Consequently, the research community lacks a clear understanding of what constitutes a robust and effective MAS framework.

Keywords— Artificial Intelligence (AI), Multi-Agent Systems (MASs), Machine Learning, IoT, Diabetes Management, Federated Learning, Mobile Health.

I. Introduction

Artificial intelligence has seen an unprecedented evolution in the last two decades beginning from simple, rule-based program that performs only one operation to advanced computational model with ability to learn, reason and take decision independently. In the context of healthcare, the development has been very revolutionary. The

early applications of AI in medicine were usually instruction-based and limited in scope, being focused only on specific problem, such as cancer screening, diagnosing tuberculosis and general classification of medical images with efficiency not exceeding 60%. However, recent advances in deep-learning, Large Language Models (LLMs) and generative AI have enabled clinicians to use these tools to solve problems of even higher complexity and magnitude. For example, now AI-based system is capable of predicting cardiovascular incidents prior to its occurrence, monitoring the progress of chronic diseases using wearable devices, analyzing complicated medical images for radiologists and assisting pharmaceutical scientists in drug discovery. Thus, these technological developments have revolutionized the whole process of practicing medicine by significantly reducing dependency on specialized clinical personnel and switching from passive treatment to pro-active approach.

Multi-Agent Systems are a system that includes a number of independent AI agents where each agent performs an assigned task and just being independent does not make them silent because once they have performed their tasks, they interact with other AI agents to achieve a collective goal. In the healthcare industry, MAS has various applications ranging from patient monitoring and vital sign recording to creating customized treatment plans. A multi-agent system framework is different from a single AI agent because while the latter tries to solve clinical problems in one step, the former allows distribution of the reasoning process among different agents, resulting in modularity and scalability that reflects the actual process of solving the problem in the clinical industry. AutoGen, LangChain, and CrewAI are some of the multiple models that have been used in developing and managing these MAS networks in healthcare, providing the structure for

classification of tasks and communication in the MAS.

The importance of this research paper is based on the fact that there is currently no standard method for evaluation of Multi-Agent Systems. While individual MAS have proved to be efficient, there are no set parameters to compare their performance.

Several critical factors are usually ignored like ethical governance, federated learning, privacy mechanisms. This research paper will address this issue by offering a systematic comparison among five latest Multi-Agent AI Systems released between 2024 and 2025 namely; IoT-MAS-T2D, HMAS-Sepsis, AgentCare, AURELIUS and EthicMAS-ICU. Comparison will be done through seven well-structured parameters that cover architecture, data handling, privacy, clinical trials, interpretability, scalability, and ethical governance. It is anticipated that the results obtained from this comparison will provide a guide for future researchers in developing Multi-Agent AI in an ethical manner within healthcare organizations around the world.

Literature Review

This literature review presents a well-organized analysis of recent Multi-Agent System-based Healthcare systems developed and implemented between 2024 and 2025, comparing their architectures, coordination and ethical considerations to extract relevant principles of designing reliable healthcare application. They include: (1) IoT-MAS-T2D Baroud (2025), an IoT integrated Multi-Agent System for controlling type-2 diabetes with wearable devices and predictive learning framework; (2) HMAS-Sepsis Borkowski & Ben-Ari (2025), seven-AI agents MAS based theoretical framework developed for sepsis management and hospital workflow optimization; (3) AgentCare (Nimbalkar et al.) (2025), Agentic AI system enabling disease detection and clinical problem-solving. (4) AURELIUS Chandrasekaran (2025), a multi-level Bayesian multi-agent augmentation learning model designed for ICU and Intensive Care environments; and, (5) EthicMAS-ICU Chen, Albarqawi & et al. (2025), GPT-4o based modular multi-agent clinical decision support system designing a particular rational AI agent.

For purposes of the current investigation, all systems will be analyzed and assessed in terms of seven criteria, namely:

(i) architecture and agent design, taking into consideration the number of agents used, synchronization procedures, layers design and

decision-making algorithms;

(ii) data input sources and compilation, which includes IoT and wearable technology connectivity, EHR parameter compatibility, and ability to process data dynamically;

(iii) privacy and security methods, which entail evaluating audit trails, regulatory frameworks, and federated learning strategies;

(iv) performance, as well as clinical efficiency and rigorousness of testing done;

(v) explainability and transparency, considering XAI tools incorporation, human-in-the-loop strategies, and clinician outputs;

(vi) scalability and deployment ability, taking into account infrastructural requirements, modularity, and applicability across other clinical fields; and

(vii) ethical and governance considerations, such as bias mitigation procedures, supervision frameworks, and governance mechanisms.

Some of the cross-cutting conclusions drawn from the comparative analysis are provided below. Differences in validation maturity can be observed. Multi-Agent System such as IoT-MAS-T2D and AURELIUS exhibit the highest validation maturity through well-executed clinical trials, while HMAS-Sepsis remains purely theoretical without any trial results that have been published. Decentralised Machine Learning turns out to become the most common privacy method in all analyzed systems, even though the degree of formalization and application of governance greatly differs. EthicMAS-ICU presents the first system where quantifiable metrics for ethics and transparency are introduced, and the selection of the escalation strategy in AURELIUS determines the level of uncertainty-aware governance. AgentCare covers the most extensive range of areas including emergency, mental health care, cancer management, and long-term illnesses, however, the validation of the system is supported by a small benchmark dataset only.

A. Competing Frameworks

There exist numerous competing models of diabetes management in conjunction with artificial intelligence (AI), each model having its own strengths and weaknesses [15, 16]. The current models vary widely ranging from standalone mobile applications to AI-driven analytics and management and support (MAS) systems [17]. Despite the ease of monitoring facilitated by mobile applications, the majority

of mobile applications lack predictive decision making in real-time. In addition, even though cloud-based centralized AI systems have predictive analytics, such systems are problematic in terms of privacy of the users' data [18, 19]. Despite the benefit that MAS systems bring in terms of improved patient care by facilitating distributed decision making, these systems still need proper privacy guarantees.

- 1) Standalone Mobile Applications (e.g. MySugr [20]): use timestamp tracking of symptoms and are limited in terms of personalized monitoring [21]
- 2) Centralized AI systems: use cloud-based augmented analytics but pose privacy risks since the data is centralized [15].
- 3) MAS systems: Kumar et al. [17] also implemented MAS to facilitate patient self-care, yet ignored FL and IoT integration.

B. Privacy Preserving Approaches

FL is an approach that tackles the issue of privacy since it enables learning on local data stored on edge devices that do not require raw data [15]. Thus, FL can be considered much more precise than differential privacy because of its accuracy while ensuring GDPR compliance [18]. It should be noted that previous MAS based approaches (e.g., [19]) did not involve FL, leading to a considerable gap in privacy-oriented diabetes treatment.

Research Gaps

There is currently no research in the area of MAS combined with FL for providing real time privacy preserving diabetes management. The study proposed aims to address this gap by incorporating Internet of Things, FL trained models and a mobile application designed specifically for patients.

II. METHODOLOGY

Methodology adopted in this project incorporates MAS in order to enhance diabetes mellitus type 2 treatment through the use of real-time monitoring, prediction and personalized recommendation capabilities.

A. System Architecture and Technical Implementation

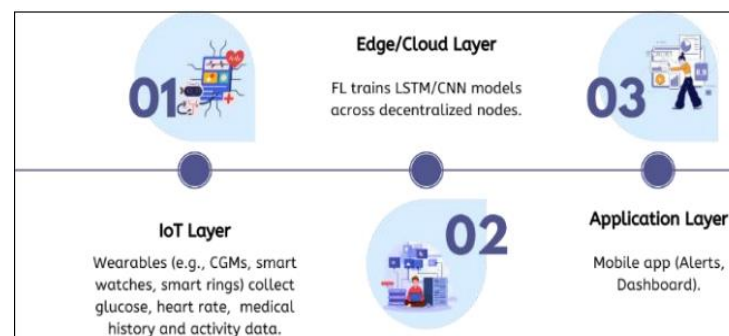
To ensure the successful integration of IoT-based devices with the cloud and AI technology to improve real-time diabetes treatment, the proposed MAS architecture can be used to enhance scalability and security and minimize latency due to edge computing and FL techniques. In this section, the description of the three-tier architecture will

highlight the contribution of the IoT devices, learning algorithms, and applications used to provide a secure and adaptive healthcare experience. The three-tier MAS includes IoT technologies, edge/cloud computing, and mobile health in which:

The IoT layer consists of wearable devices such as CGM sensors, smart watches, rings, and bands that provide information about glucose level, heart rate, physical activities, and personal health records of the patient. Edge computing is performed by the edge device for preprocessing data.

Edge/cloud layer: LSTM/CNN models are used for FL.

Mobile application: Developed based on the Flutter framework.



Sepsis Management by Andrew A. Borkowski:

Despite numerous advances being witnessed in broad spectrum antibiotics, medical imaging and life-support equipment technologies, deaths from sepsis remain at an extremely high rate. There have been challenges in achieving clinical optimization in relation to sepsis management. Predicting models for sepsis patients have proved to be quite challenging in the past. This paper outlines a multi-agent artificial intelligence model aimed at improving the process of patient monitoring.

Data Collection and Integration Agent. In collaboration with the controlled vocabulary that will describe all types of data, the first step to be taken by the data collection and integration agent involves cleansing, transforming, and arranging patient data in structured and unstructured formats. The data collection and integration agent makes summary statements of the consultants' notes. The agent arranges the data in a manner that can be accessed by both humans

and machine systems. All numeric data is presented graphically including any trends over time in the past. In addition, all the orders are captured in the form of digital data by use of the controlled vocabulary. Besides, the data structures provided by these sources are useful in training the agents in the future. **Diagnostic Agent.** The presence of critical illness involves multiple pathologies and abnormalities observed in different kinds of studies, including chest X-ray studies, CT scans, cellular components of blood, chemical composition of plasma, microscopic studies of tissues and so on. Moreover, data from life support parameters helps in understanding the severity of the patient's disease status and helps in decision-making in management. There is an abundance of data available in terms of visualization as well as numerical data which can be employed in calculations and recommendations as well as in training. For example, in case of fluid overload evaluation in chest X-rays or in histopathological sections, AI agents employ deep learning models such as convolutional neural networks and vision transformers. Time series data, such as vital signs data, is analyzed by employing RNNs and transformers.

Risk Stratification Agent: It assesses the seriousness and forecasts the probable results of each case in question based on probabilities of morbidity and mortality. Such probabilities are calculated according to an already existing scoring system and adjusted with respect to the past histories of patients in relation to other agents' conditional patients. This is depicted in charts where the key risks are indicated. **Treatment Recommendation Agent.** The treatment recommendation agent uses reinforcement learning and updated clinical guidelines that consider the data on the same cases from the past. The data in question is structured within the vocabulary of similar patients. The training process is conducted using data on the physiological features of the patient in question. Recommendations are formulated in the form of a special user interface with recommendations for orderable items and references to articles from the literature review of previous research. Stop rules are used for ceasing computing if recommendations are unclear or if a certain pathway cannot be computed reliably, prompting human intervention.

Resource Management Agent: This agent manages resources of a hospital by using constraint programming techniques, queue theories for simulating patient flow and genetic algorithms for solving difficult scheduling problems.

Monitoring and Alert Agent: This agent observes patient conditions and raises alerts whenever

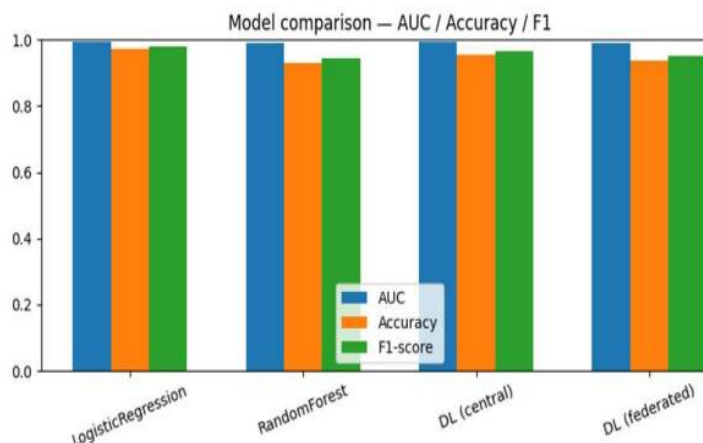
changes in their condition are observed. This monitoring and alert agent utilizes anomaly detection algorithms to find out anomalies in data pertaining to patients and implements time-series forecasting algorithms such as ARIMA and Prophet for predicting patients' conditions. Additionally, stream-processing algorithms can be utilized for real-time data analysis.

Documentation and Reporting Agent. This agent will document the medical records of all patients, perform reporting, and implement several other strategies for achieving this purpose. These include natural language processing for generating reports, advanced language learning models, specialized for medical corpus for creating narratives, and information retrieval for efficient querying.

Multi Agent Framework for Real-Time Disease Detection and Patient Support:

Thus, this proposed solution has the potential of making substantial contributions to this domain owing to several novel features. First of all, the present research project encompasses development of an entirely new multi-agent AI framework which includes both real-time disease detection as well as patients' support. Furthermore, this approach is built around the principle of federated learning and utilizes federated learning technology for privacy-preserving collaboration between multiple healthcare organizations. This approach also allows integrating disease diagnosis and real-time patient assistance with explainability and decision transparency in order to develop a next-generation healthcare solution. The application potential of the method will be illustrated by various examples from the healthcare field, including disease prevention, chronic diseases management, cancer diagnostics, and emergencies treatment. To give a better understanding of the research problem and the proposed method, the contribution of this work can be summarized as follows. Firstly, it develops a new multi-agent AI system for disease diagnosis and real-time patient assistance. Secondly, it makes use of federated learning technology as a platform for collaboration between healthcare organizations while providing patients with privacy guarantees. Thirdly, it incorporates decision explainability that guarantees transparency of the decision-making process in a machine learning model. Finally, this AI model can be

employed in multiple healthcare applications.



AURELIUS Multi-agents Unifying System for Addressing Non-determinism within HealthTech:

Due to its design, AURELIUS can be used as a multi-agent system that integrates probabilistic inference, collaboration between agents, and hierarchical reinforcement learning as a highly sophisticated decision-making tool in HealthTech applications. Being developed according to the principles of Bayesian statistics, it correctly distinguishes between aleatoric and epistemic uncertainties, ensuring that reliable confidence intervals could be calculated. In addition, due to the use of the Dec-POMDP framework, the representation of a partially observable and decentralized healthcare environment becomes more accurate, while hierarchical MARL increases sample efficiency and temporal abstraction. The structure of AURELIUS is characterized by three main elements, such as: (i) multimodal Bayesian encoders which integrate diverse kinds of patients' data with consideration of uncertainty; (ii) interagent graph communication for the purposes of collaborating actions; and (iii) selective escalation in case of high uncertainty, leaving some tasks to be solved manually. In summary, the presented characteristics provide such benefits as high reliability, clear explanation, and compliance. In contrast to standard MARL approaches that seek to maximize the gains of agents, AURELIUS implements concepts of governance and safety by imposing uncertainty penalties and abstinence control. In more detail, recent innovations in the development of multimodal agents demonstrate the need for creating efficient architectures for rapid prototyping [16].

Improving Clinical Decisions:

According to the results of our research, MAS forms SAS concerning the accuracy of

forecasting the mortality rate, where MAS and SAS have 59% and 56%, respectively. In addition, the LOS of MAS is lower compared to SAS while being equally transparent. Based on the outcomes of our experiment, transparency of MAS equals 85.5%. The transparency of SAS is even higher, and it makes 86.21%. Consequently, it is possible to state that our MAS manages to achieve satisfactory transparency regardless of the complexity of coordination of specialized agents.

The relationship between the efficiency of the system and the transparency of decision-making becomes an important consideration while creating DSSs. Given the necessity of precise predictions, the use of the MAS approach turns out to be extremely valuable. Being a system that ensures adequate transparency, the MAS system makes decision making more efficient and, therefore, helps satisfy both needs for producing the desired results and building up some justification of particular decisions in order to use them in the medical practice. But when transparency of reasoning behind predictions plays the key role in terms of earning the trust of clinicians, there is still much to do with regard to the SAS area.

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